Aryan Kumar

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SKILLS

3D modelling and Simulation
- SolidWorks
- Fusion360
- AutoCad
- Ansys Workbench

CERTIFICATIONS

- Certified Solidworks Associate (CSWA)
- AutoCad 2D & 3D (Udemy)
- Generative Design in Fusion360

EDUCATION

B.Tech Mechanical Engineering
NIT Tiruchirappalli
2023 – present

Senior Secondary Education
St. Dominic Savio's High School
2021 – 2023

RESEARCH INTERNSHIP

Evaluating Whole Body Vibration Effects on the Lumbar Spine using FEM
IIT Hyderabad
05/2025
- Developed a detailed finite element model of the **human lumbar spine** based on MRI data and **50th percentile male** including vertebrae and intervertebral discs.
- Conducted **static** and **dynamic** simulations to analyse **biomechanical behaviour**, performing range of motion, intradiscal pressure, and vibration analyses to validate the model and study lumbar spine responses under various loading and whole-body vibration conditions.
- Provided insights into lumbar spine injury mechanisms under vibration by identifying stress and strain patterns in the lumbar discs, highlighting **regions prone to damage** and offering information valuable for improving **driver health and ergonomic design**.

PROJECTS

SILICO DAMP
Low Noise Motorcycle Stand Dampener
03/2025
- Designed and simulated **a silicone-based dampening solution** using **CAD/CAE** tools to reduce noise from motorcycle stand impacts by increasing strain energy absorption.
- Conducted cost-benefit and manufacturability analysis comparing standard and **high-damping silicone rubber pads**; proposed **compression molding** for scalable production.
- Recommended process optimizations including injection molding, mold redesign to **reduce manufacturing costs by ~20%** while maintaining product performance.

MOTOPULSE
Electric Motor Conversion for Normal cycles
02/2025
- Designed a **36V, 250W BLDC mid drive motor-based** system in **Solidworks** to convert regular bicycles into electric-powered ones, making them more efficient and accessible.
- Engineered a **Custom Spindle and Shaft Lock Mechanism** by extending the main spindle and adding two hollow shafts for the pedals, allowing the right pedal to remain independent of the main crank. A **ball bearing system** inside the hollow shaft enabled smooth movement, while a **push-pin lock mechanism.**
- Designed a system where riders could **switch between electric and manual pedalling** by simply locking or unlocking the spindle, ensuring smooth power delivery, better efficiency, and a natural riding experience.

ACHIEVEMENTS

- **Bajaj TORQ '25 National Finalists**